PROBLEM OF THE MONTH

DEPARTMENT OF MATHEMTICS
ELMHURST UNIVERSITY
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Problem. Let $a_1, a_2, \ldots, a_k$ be real numbers, and let $m_1, m_2, \ldots, m_k$ be positive integers. Consider the function

$$f(x) = a_1 \sin(m_1 x) + a_2 \sin(m_2 x) + \cdots + a_k \sin(m_k x).$$

Show that f has a real root.

Scan the following QR code for submission instructions.